

AIDATACENTER — SOURCE OF TRUTH

NetBox as Source of Truth

Rev 0.1 · 2026-07-18

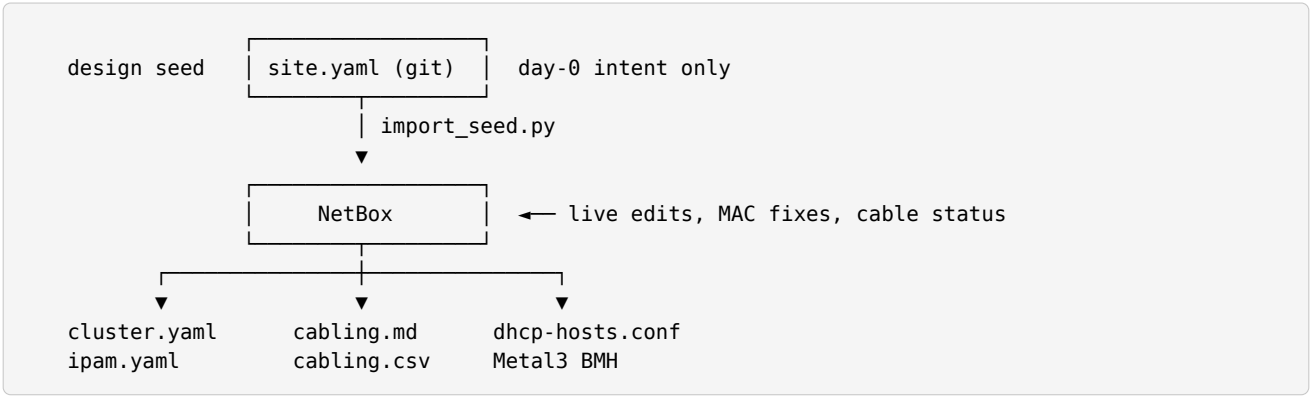
Generated from Markdown — edit the .md source and regenerate (python3 docs/md2pdf.py)

Contents

1 Day-0 (before K8s)	3
2 Day-2 (production NetBox)	4
3 Custom fields	5
4 IPAM plan (baseline)	6
5 Change control	7
6 RBAC (suggested)	8
7 Integration checklist	9
8 Related	10

NetBox is the **authoritative DCIM + IPAM** system for this cluster. Inventory, prefixes, interfaces, and cables are not maintained long-term in Git YAML (except design seed pre-go-live).

Concern	SoT	Export consumer
Devices, roles, racks	NetBox DCIM	export_inventory.py → Metal3 BMH / seed
Mgmt + OOB + VIP IPs	NetBox IPAM	inventory + dnsmasq
Prefixes / VLANs	NetBox IPAM	inventory/ipam.yaml, network automation later
Host↔leaf, leaf↔spine, BMC cables	NetBox cables	docs/cabling.md + install CSV
BMC credentials	Vault / env	never in NetBox secrets by default
K8s desired state	Git (Flux)	lattice of CRs



1 Day-0 (before K8s)

On seed01:

```
cd bootstrap/netbox
cp env.example .env && $EDITOR .env # SECRET_KEY + API_TOKEN_PEPPER_1 required
docker compose up -d
export NETBOX_URL=http://127.0.0.1:8081
export NETBOX_TOKEN=$(bash scripts/create_token.sh) # v2 token: nbt_<key>.<secret>
pip install -r requirements.txt
python3 scripts/import_seed.py
bash ../scripts/netbox-sync.sh
```

Without Docker (design-time / CI):

```
bash bootstrap/scripts/netbox-sync.sh --offline
```

Offline expands seed/site.yaml (same rail math as import).

NetBox 4.6 notes: API tokens are v2 (nbt_..., HMAC-signed; plaintext shown once at creation — API_TOKEN_PEPPER_1 must be set). MACs are discrete `MACAddress` objects marked primary per interface. Prefixes use generic scope (site), VLANs live in a site-scoped VLAN group (direct site assignment deprecated).

2 Day-2 (production NetBox)

Flux path: `bootstrap/platform/apps/netbox` (`platform-netbox` Kustomization, chart 8.x / NetBox v4.6.5).

After URL is `https://netbox.ai.local`:

1. Create API token (Vault: `secret/netbox/api_token`) — v2 format `nbt_<key>.<secret>`; the chart auto-generates signing peppers.
2. Re-run `import_seed.py` once **only if** the K8s instance is empty (or use DB dump from seed NetBox).
3. Point automation `NETBOX_URL` at cluster NetBox.
4. Decommission seed Docker NetBox after cutover.

Preferred cutover: `pg_dump` seed Postgres → restore into Helm Postgres PVC, preserving IDs and cable history.

3 Custom fields

Field	Type	On	Use
k8s_role	text	device	control-plane, gpu-worker, ...
gpu_count	int	device	8 on workers
rail_index	int	leaf	0-7
redfish_url	text	device	Metal3 BMC address

4 IPAM plan (baseline)

Prefix	VLAN	Role
10.10.0.0/24	10 mgmt	OS, CP, VIP, seed
10.20.0.0/24	20 oob	BMC + switch Management1
10.30.0.0/16	fabric	Underlay container (expand later)
10.40.0.0/16	storage	STOR data
10.96.0.0/12	—	K8s services (documented only)
10.244.0.0/16	—	K8s pods (documented only)

API VIP: **10.10.0.20** → **api.ai.local**.

5 Change control

Change	Where	Next step
New MAC after RMA	NetBox interface	<code>netbox-sync.sh</code> → re-render BMH
New mgmt IP	NetBox IPAM assign	<code>sync</code> → <code>dnsmasq</code> + BMH
Cable installed	Cable status → connected	re-export cabling guide
Add GPU rack	NetBox rack + devices + cables	update seed only if design template changes
Fabric design change	Update <code>seed/site.yaml</code> <code>sabbling_policy</code> or edit cables in UI	<code>sync</code>

Git PR is still required for: CAPI templates, Flux apps, seed schema versions. **Not** for everyday MAC/IP/cable status.

6 RBAC (suggested)

Group	NetBox perms
netops	DCIM + cables + IPAM write
compute	devices read, interfaces write (MAC)
platform	read-all + cluster objects
audit	read-only

7 Integration checklist

- ☐ Seed NetBox healthy; import finishes without missing ifaces
 - ☐ `cluster.yaml` header contains **GENERATED**
 - ☐ Cable counts: 64 host + 64 leaf-spine + BMC count match export
 - ☐ CAPI `02-apply-cluster.sh` consumes exported inventory
 - ☐ Production NetBox under Flux; seed retired or read-only
-

8 Related

- [bootstrap/netbox/README.md](#)
- [docs/cabling.md](#)
- [docs/bootstrap.md](#)